

Infosys Springboard Virtual Internship 6.0 Completion Report

Team Details

Batch Number: 3
Start date: 22 Sep 2025
Name: U Surya Sasikanth
Internship Duration: 8 Weeks

1. Project Title: CrowdCount-People Counting Using Video Analytics

2. Project Objective: The main objective of this project is to design and develop an intelligent video analytics system that automatically detects, tracks, and counts people in real time across different zones using advanced computer vision techniques. The system aims to provide accurate crowd density estimation and live zone-based statistics to assist in effective crowd management and safety monitoring.

This project is intended to support event organizers, security teams, and urban planners by offering real-time insights into crowd flow and occupancy levels. By leveraging object detection and tracking algorithms integrated with a web-based dashboard, the system delivers both analytical visualization and interactive tools for monitoring, alerting, and decision-making in dynamic environments.

3. Project Description in Detail: CrowdCount: People Counting Using Video Analytics is a Python-based application built using the Flask web framework and integrated with state-of-the-art computer vision models. It is designed to automate real-time crowd detection, tracking, and counting processes from live camera feeds or uploaded videos, combining deep learning techniques with interactive visualization and alerting functionalities.

Approach Taken:

1. Requirement Analysis: The project began with an in-depth study of the challenges faced by security operators and event organizers — such as the need for real-time crowd monitoring, difficulties in estimating zone occupancy manually, and the lack of integrated systems for people counting and density alerts.

Requirements were gathered to ensure the system could stream live camera feeds, support multi-zone detection, track unique individuals without duplication, and generate visual dashboards for real-time insights and alerts.

2. System Design and Architecture: A modular and scalable architecture was designed to separate the detection, tracking, analytics, and visualization layers of the system.

The architecture includes:

- A video ingestion module for streaming webcam or IP camera feeds.
- An AI processing layer integrating YOLOv8 and Deep SORT for detection and unique ID tracking.
- A Flask-based web interface for user and admin interactions.
- A real-time analytics dashboard powered by Chart.js for displaying live counts, heatmaps, and alerts.
- Role-based access control and logging modules for secure and transparent operations.
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3.Development: Implemented real-time video pipelines to capture frames, detect human presence, and map movements across defined zones.

- Optimized YOLOv8 models for efficient inference and accuracy in varied lighting and crowd conditions.
- Developed a tracking service that assigns persistent IDs to each person to avoid double counting.
- Integrated an alert system that triggers notifications when zone thresholds are exceeded.

Designed an interactive Flask dashboard featuring:

- Zone-based people count visualization.
- Live bubble and line charts using Chart.js.
- Alert banners for threshold violations.
- Admin panel for managing users, cameras, and reports.
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4.Testing and Deployment: Comprehensive testing was carried out, including:

- Functional testing to validate detection, tracking, and counting modules.
- Performance testing to ensure stable operation under multiple video streams.
- Accuracy evaluation comparing model predictions against manually counted data.
- Usability testing of the dashboard for responsiveness, clarity, and ease of use.

Impact and Real-World Implementation

- **Crowd Management and Safety:** Enables real-time monitoring of crowd density and movement to prevent overcrowding and ensure public safety.
- **Security and Operations:** Supports authorities with data-driven insights for entry control, evacuation planning, and resource allocation.
- **Research and Planning:** Provides useful data for studying crowd behavior and improving space utilization in smart city environments.
- **Scalability:** Can be extended across multiple cameras and IoT systems for large-scale deployment and predictive analytics.

4. Timeline Overview

Week	Activities Planned	Activities Completed
Week 1	Introduction to project objectives, environment setup, and basic Python practice.	Project initiated, environment configured, and foundational Python programs completed.
Week 2	Implement user authentication and registration module (Milestone 1 start).	Login and registration system developed using Flask; database connection verified.
Week 3	Learn libraries – NumPy, Pandas, and visualization tools; set up JWT security.	Data handling modules integrated; JWT-based authentication and secure sessions added.
Week 4	Develop camera streaming and zone drawing (Milestone 2).	Live webcam/video streaming implemented; interactive zone creation and editing enabled.
Week 5	Integrate YOLOv8 + Deep SORT for detection and tracking (Milestone 3).	Object detection and tracking functional with >90% accuracy; zone-wise counts updated in real time.
Week 6	Build dashboard with live charts, alerts, and export features (Milestone 4).	Dashboard designed with Chart.js; alert system, CSV/PDF exports, and analytics completed.
Week 7	Integrate all milestones and perform full system testing.	All modules combined; live tracking and admin analytics successfully demonstrated.
Week 8	Final testing, documentation, and project presentation.	Testing finalized, report prepared, and final demo presented successfully.

5a.KeyMilestones

Milestone	Description	Date Achieved
Project Kickoff	Defined project objectives, scope, and collected initial video datasets for detection and tracking analysis.	Sep 25, 2025
Prototype / First Draft	Developed the initial Flask setup and zone drawing module; implemented basic webcam streaming and media upload features.	Oct 8, 2025
Mid-Term Review	Presented progress on people detection and tracking integration using YOLOv8 and Deep SORT; discussed accuracy improvements and received mentor feedback.	Oct 16, 2025
Final Submission	Completed full system development including alert mechanism, admin dashboard, and user role management; finalized documentation.	Nov 6, 2025
Presentation	Demonstrated the complete CrowdCount system, showcasing live crowd tracking, real-time analytics dashboard, and alert functionality.	Nov 12, 2025

5b. Project execution details

The CrowdCount: People Counting Using Video Analytics system uses computer vision and deep learning to detect, track, and count people in real time. It offers a web-based dashboard for live monitoring, analytics, and alerts.

Planning & Design: Defined objectives, gathered requirements, and designed a modular architecture using Flask, YOLOv8, and Chart.js.

Development & Integration: Implemented live video streaming, zone drawing, and detection modules. Integrated tracking, alert logic, and secure user authentication.

Dashboard & Visualization: Built an interactive dashboard to display real-time counts, zone statistics, and alerts. Added admin features for camera setup and data export.

Testing & Deployment: Tested system accuracy, stability, and dashboard performance. Deployed the final application on a local Flask server for demonstration.

Enhancements: Included analytics visualization, CSV/PDF exports, and final documentation for project submission.

Project Milestones:-

1 Milestone 1 — Initial Setup

- Stream camera successfully.
- Save and render zones on live feed.
- Interface for zone editing functional.

2 Milestone 2 — Detection & Tracking

- Detection and tracking with over 90% accuracy.
- Unique people ID tracking across frames.
- Real-time zone-based counting updates.

3 Milestone 3 — Visualization and Alerts

- Dashboard shows updated statistics and live heatmap.
- Export features and alert notifications functional.
- Visuals are responsive and easy to interpret.

4 Milestone 4 — Administration & Security

- Admin can control all camera settings and manage reports.
- Secure role-based access system operational.
- Complete system logs and analytics accessible.

Tech Stack:-

Category	Tools & Technologies Used
Programming Language	Python, JavaScript
Backend Framework	Flask
Frontend Technologies	HTML5, CSS3, Chart.js, Canvas API
Machine Learning / AI Model	YOLOv8 (Object Detection), Deep SORT (Tracking)
Database	SQLite3
Visualization & UI	Chart.js, Flask Templates (Jinja2)
Libraries & Packages	OpenCV, Ultralytics, FPDF, JWT, Werkzeug
Version Control & Deployment	Git, Render / Local Server
Environment	Python Virtual Environment (venv)

6. Snapshots / Screenshots

Fig-1: Login Page

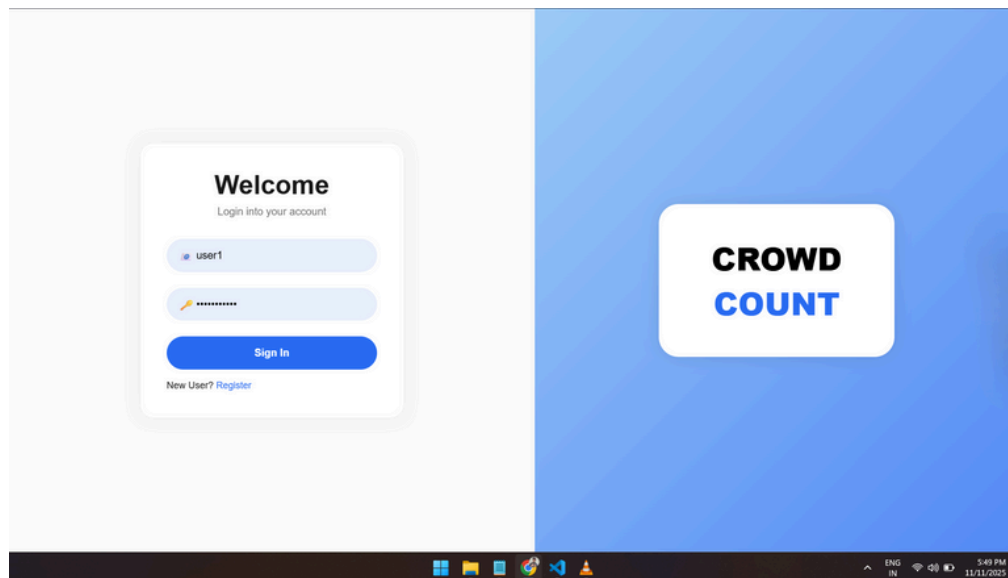


Fig-2: Registration Page

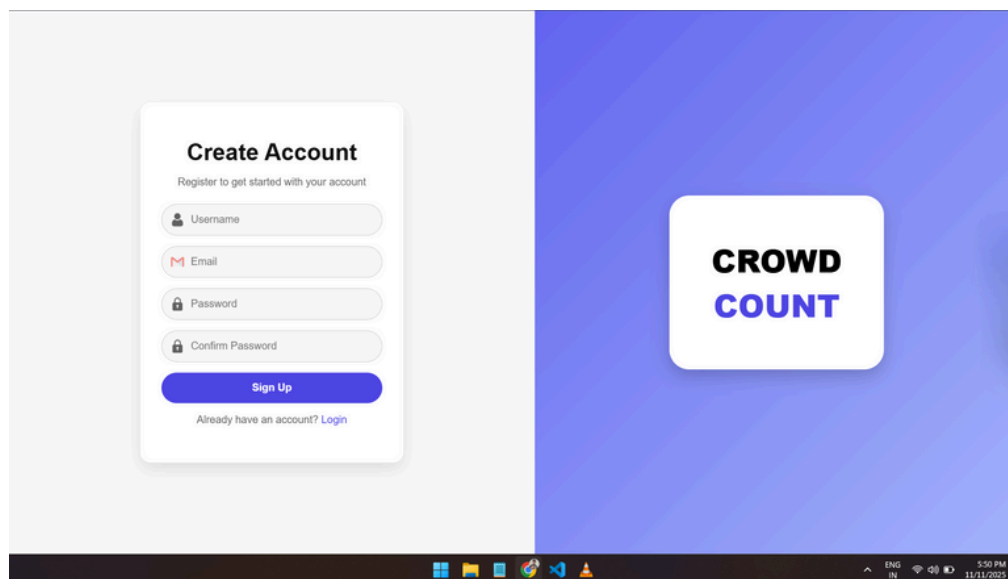


Fig-3: Home Dashboard

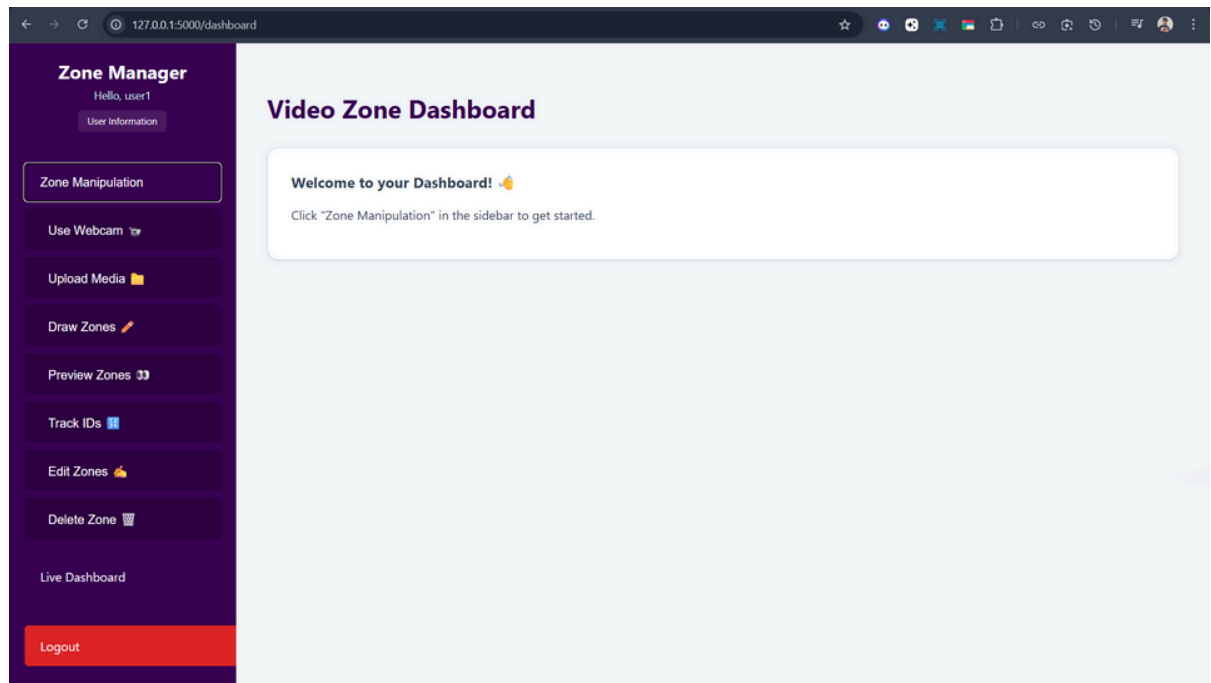


Fig-4: Profile Page

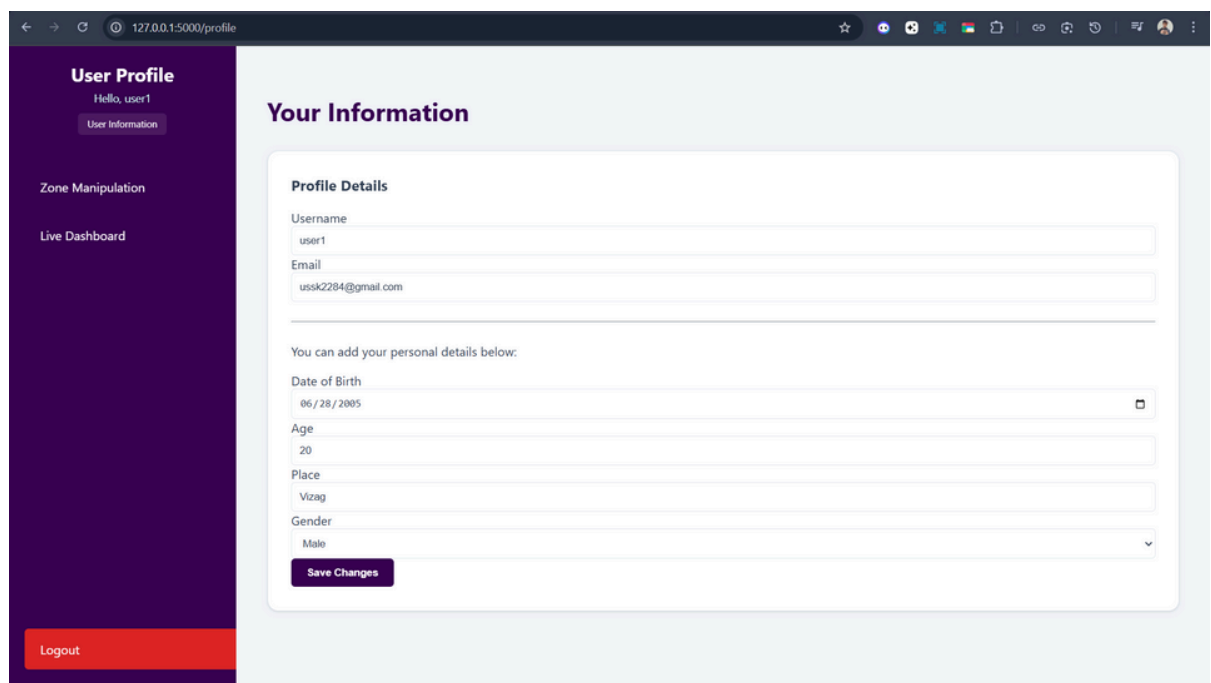


Fig-5: Media Uploade here

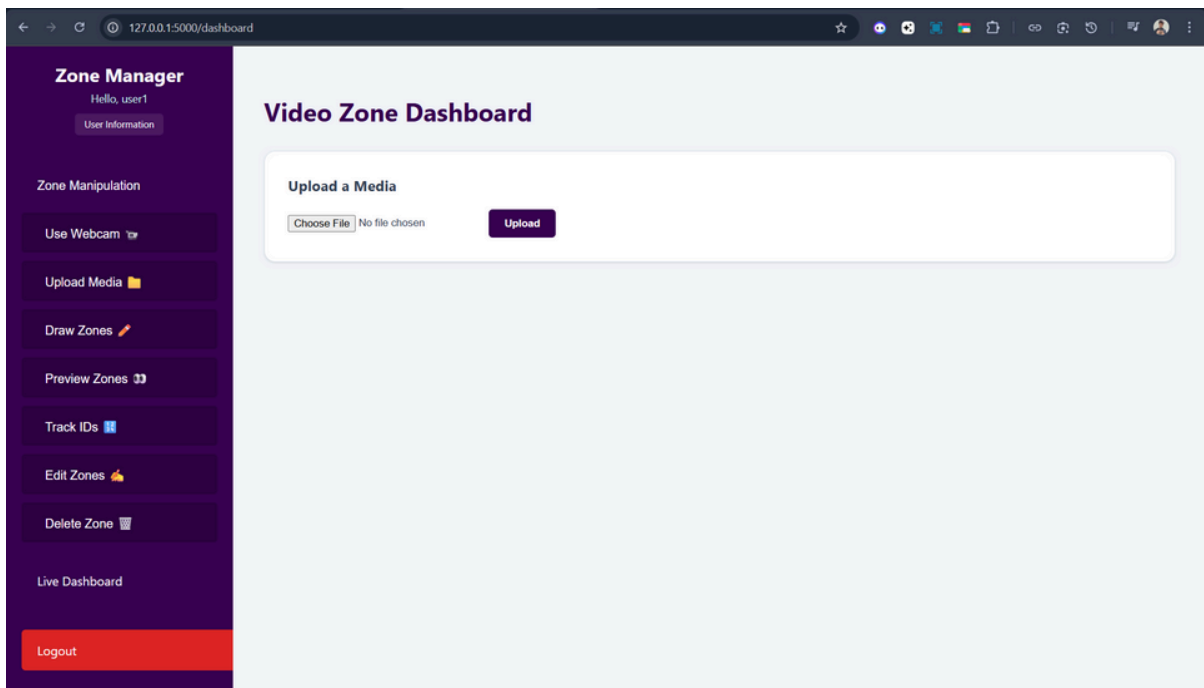


Fig-6: Zone Drawing Tool

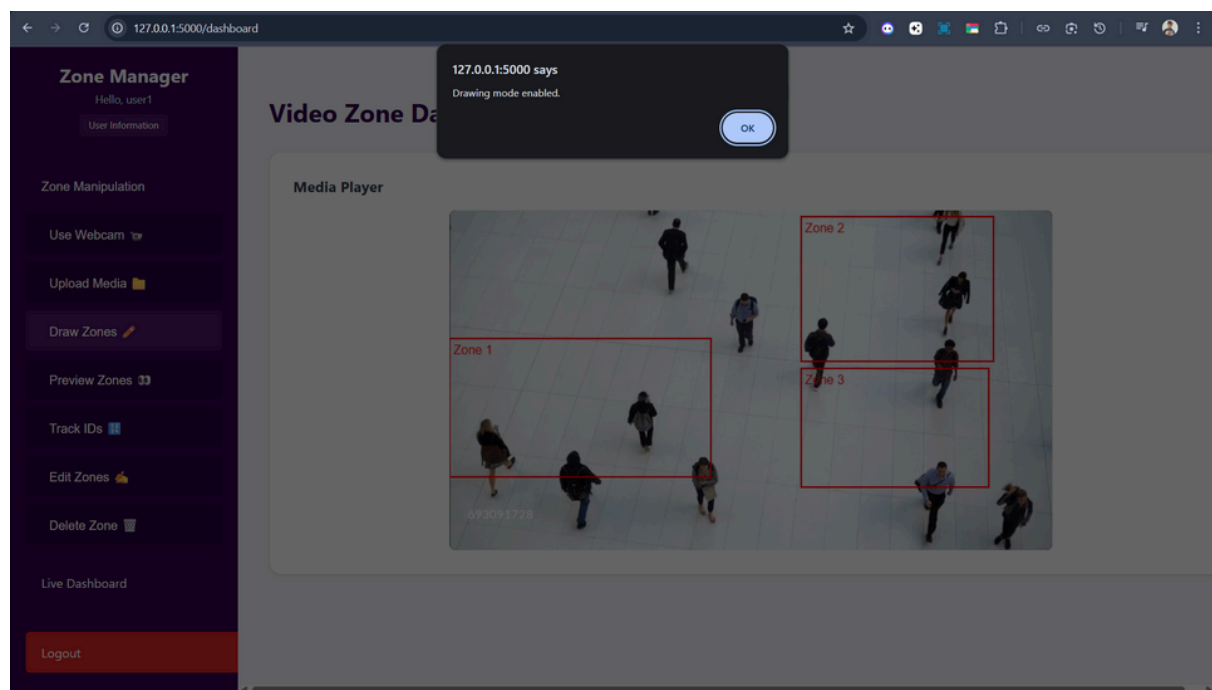


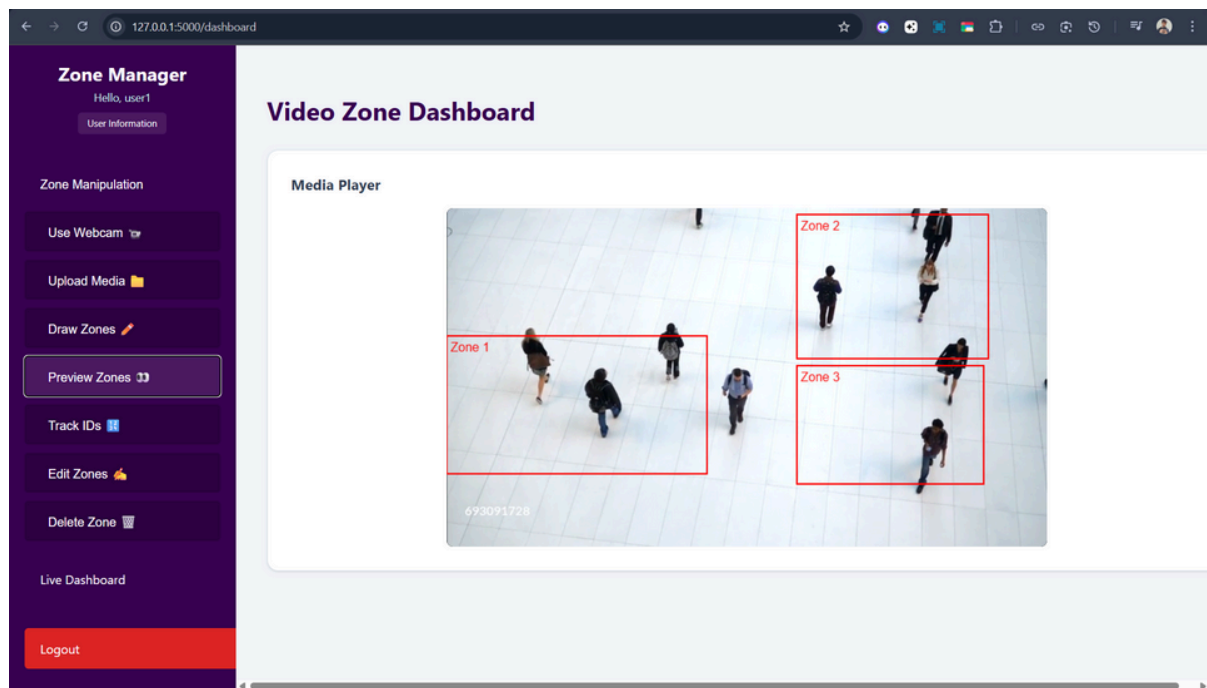
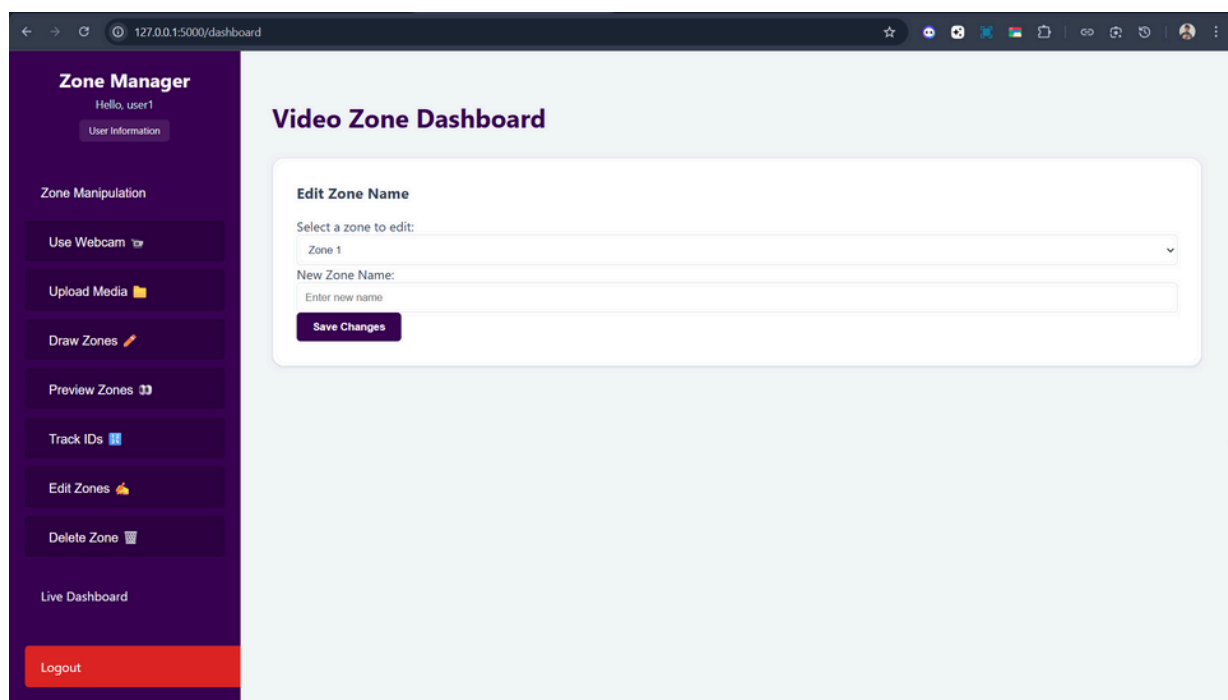
Fig-7: Preview Zones Page**Fig-8: Edit (Rename) Zones**

Fig-9: Delete Zones Page

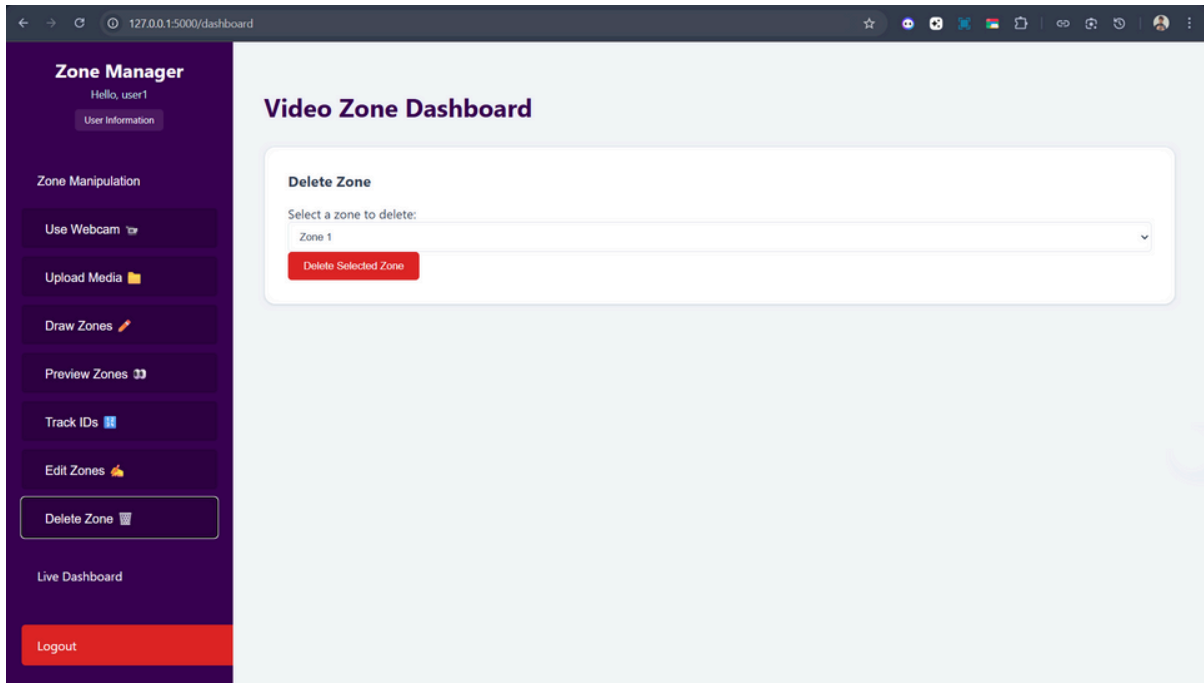


Fig-10: Live Dashboard

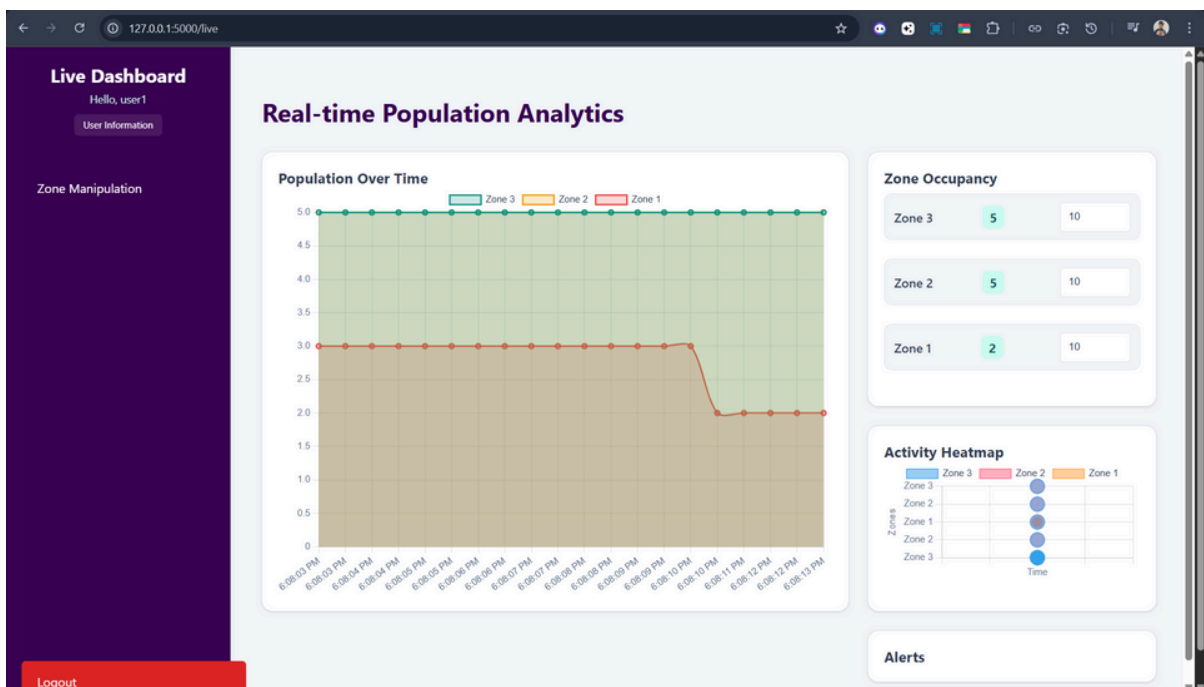


Fig-11: Alerts by Threshold

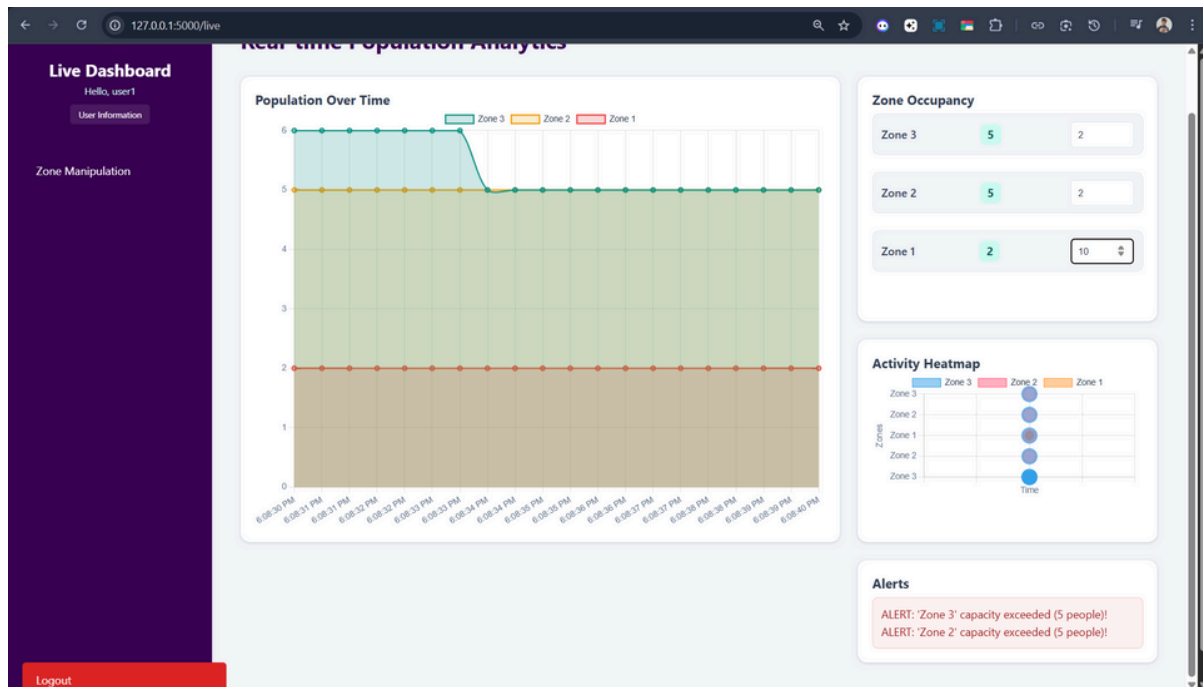


Fig-12: Admin Dashboard

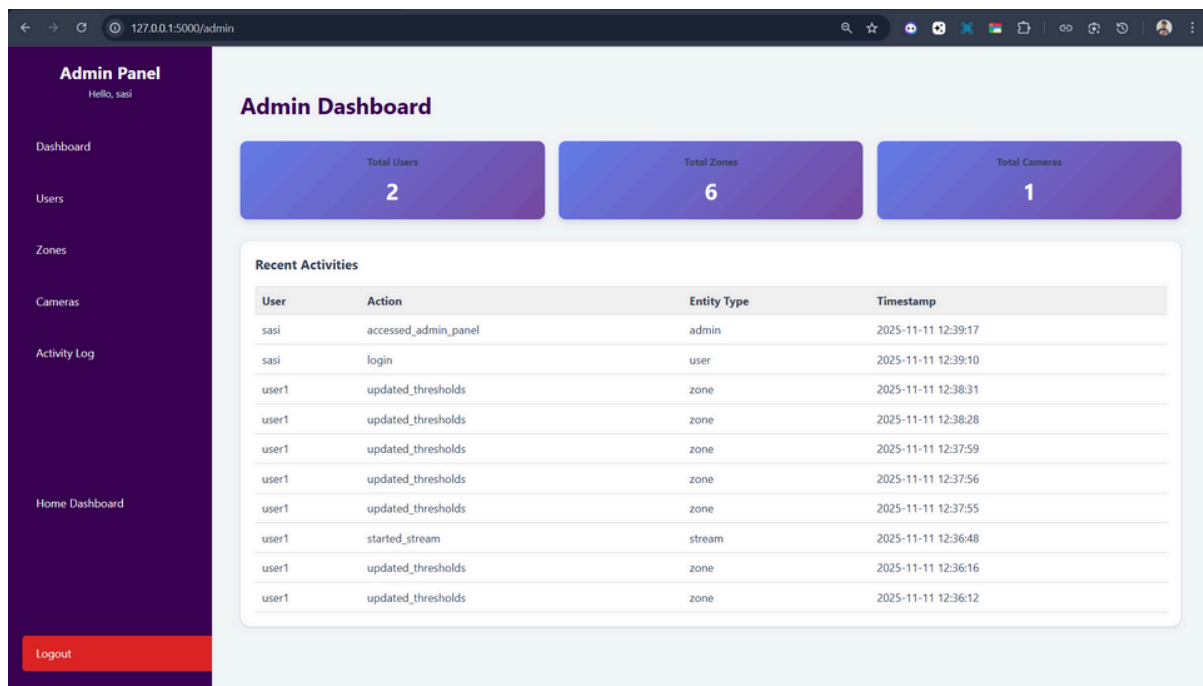


Fig-13: Streamlit Web Dashboard

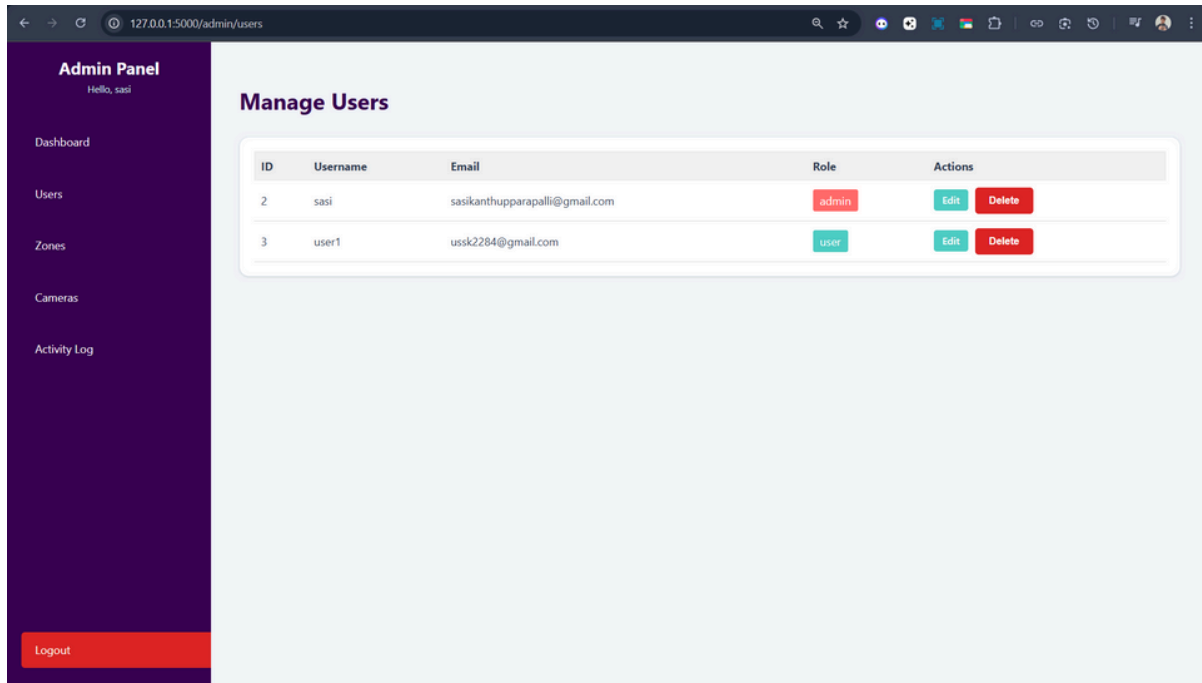


Fig-14: Users in Admin Pannel

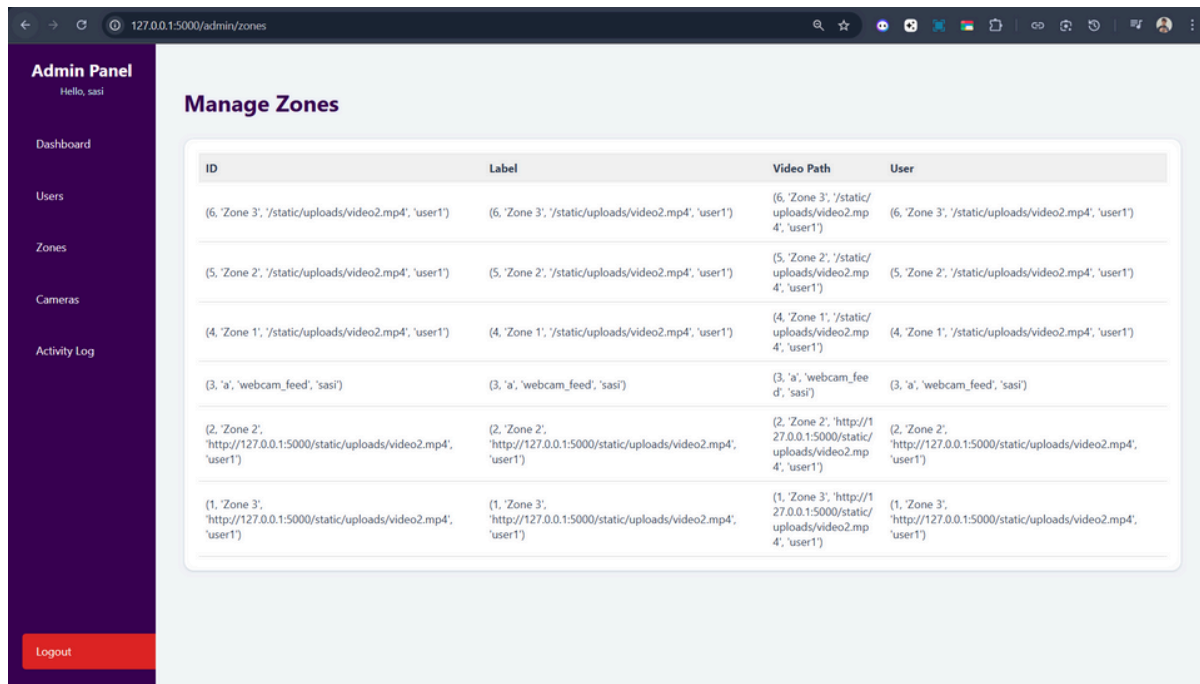


Fig-15: Manage Cameras

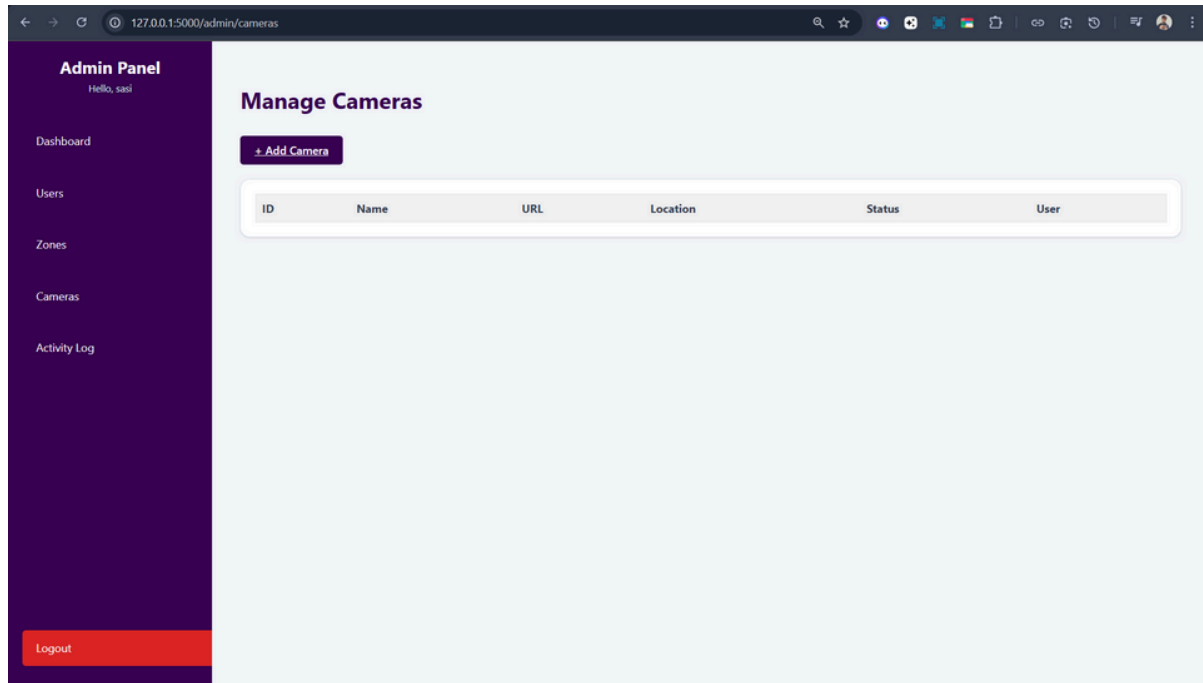
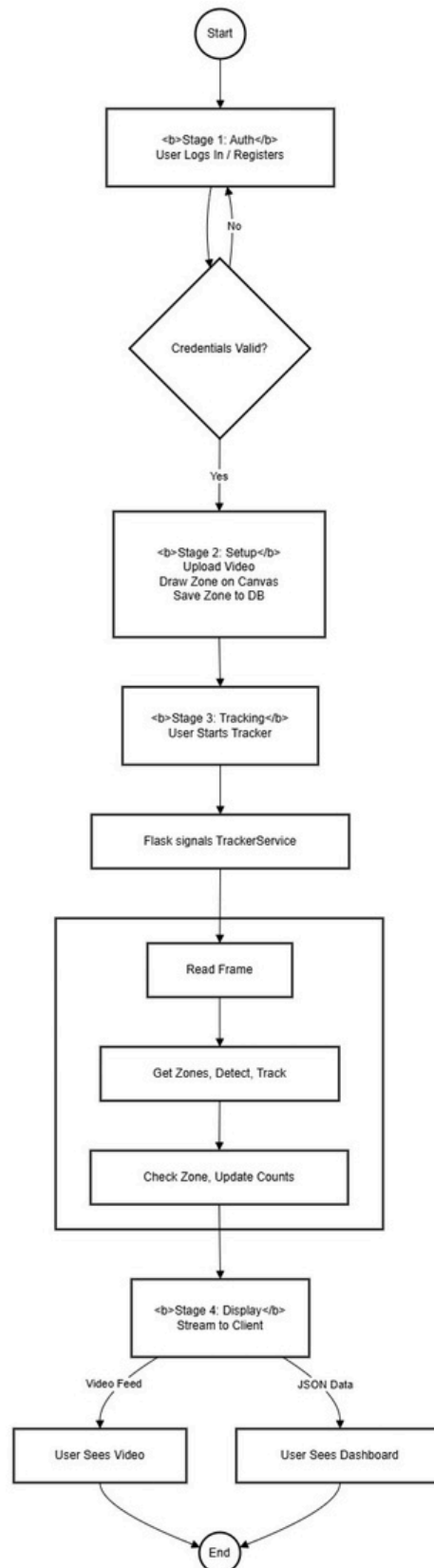


Fig-16: Active Log Data To export

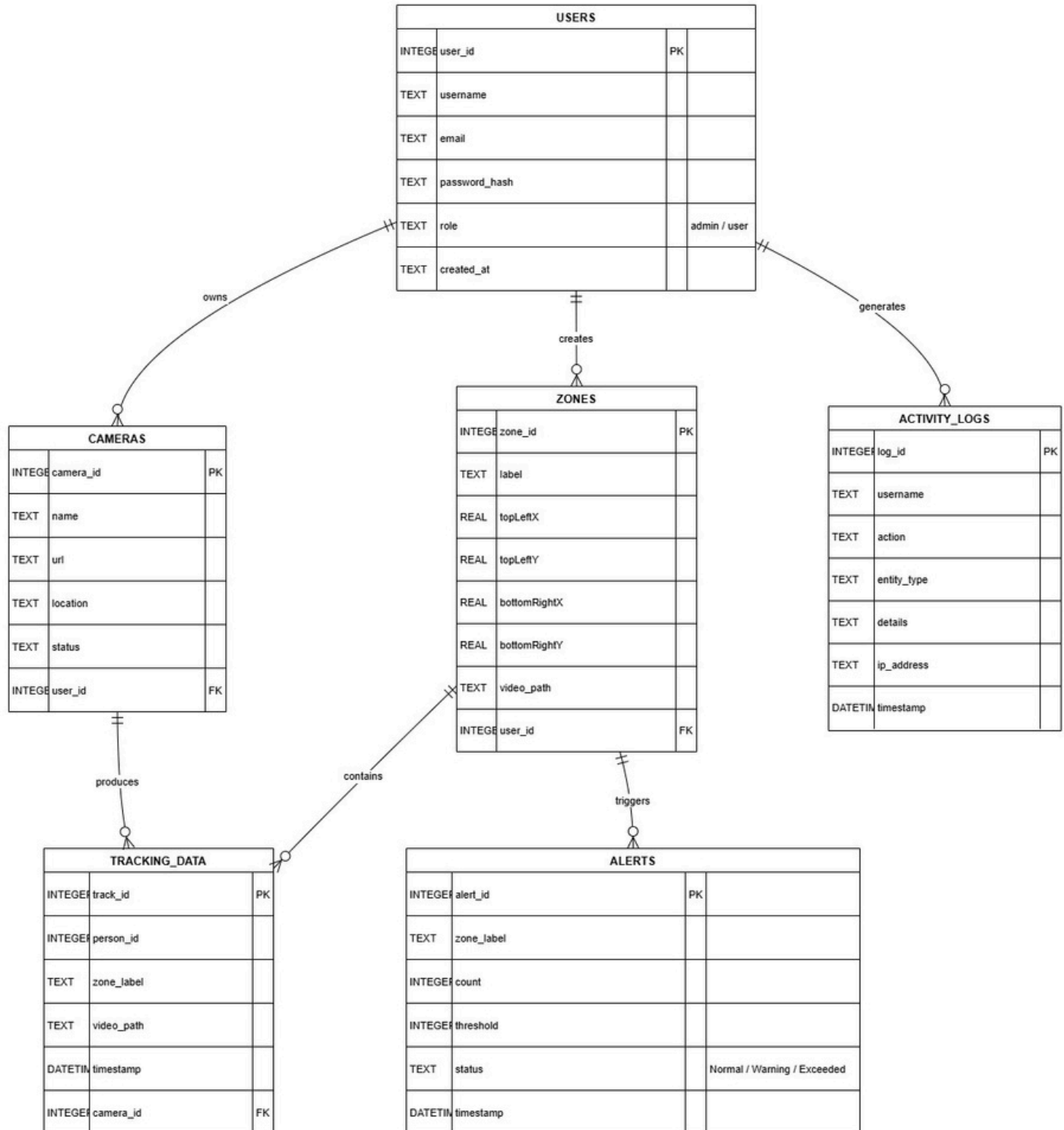
The screenshot shows the 'Activity Logs' interface. On the left is a dark purple sidebar with the 'Admin Panel' header and a 'Logout' button at the bottom. The sidebar menu includes 'Dashboard', 'Users', 'Zones', 'Cameras', and 'Activity Log'. The main area is titled 'Activity Logs' and features 'Export CSV' and 'Export PDF' buttons. Below these is a table with the following columns: ID, Username, Action, Entity Type, Details, IP Address, and Timestamp.

ID	Username	Action	Entity Type	Details	IP Address	Timestamp
363	sasi	viewed_cameras	admin	Viewed all cameras	127.0.0.1	2025-11-11 12:40:21
362	sasi	viewed_zones	admin	Viewed all zones	127.0.0.1	2025-11-11 12:39:58
361	sasi	viewed_users	admin	Viewed users list	127.0.0.1	2025-11-11 12:39:35
360	sasi	accessed_admin_panel	admin	Accessed admin dashboard	127.0.0.1	2025-11-11 12:39:19
359	sasi	accessed_admin_panel	admin	Accessed admin dashboard	127.0.0.1	2025-11-11 12:39:17
358	sasi	login	user	User logged in	127.0.0.1	2025-11-11 12:39:10
357	user1	updated_thresholds	zone	Updated zone thresholds	127.0.0.1	2025-11-11 12:38:31
356	user1	updated_thresholds	zone	Updated zone thresholds	127.0.0.1	2025-11-11 12:38:28
355	user1	updated_thresholds	zone	Updated zone thresholds	127.0.0.1	2025-11-11 12:37:59
354	user1	updated_thresholds	zone	Updated zone thresholds	127.0.0.1	2025-11-11 12:37:56
353	user1	updated_thresholds	zone	Updated zone thresholds	127.0.0.1	2025-11-11 12:37:55
352	user1	started_stream	stream	Started live stream	127.0.0.1	2025-11-11 12:36:48
351	user1	updated_thresholds	zone	Updated zone thresholds	127.0.0.1	2025-11-11 12:36:16
350	user1	updated_thresholds	zone	Updated zone thresholds	127.0.0.1	2025-11-11 12:36:12
349	user1	updated_thresholds	zone	Updated zone thresholds	127.0.0.1	2025-11-11 12:35:57

Workflow Diagram:-



Database Schema:-



7. Challenges Faced

During the project, various challenges emerged across technical, operational, and integration stages. Each presented unique opportunities for skill enhancement and problem-solving, contributing to the system's overall robustness.

1. Technical Challenges

- **Model Performance:** Achieving consistent detection accuracy across lighting and crowd density variations was initially difficult.
- **Resolution:** Enhanced YOLOv8 model with adaptive confidence thresholds and tested on diverse datasets to stabilize performance.
- **Real-Time Processing:** Maintaining smooth frame rates during live detection caused latency on lower-end systems.
- **Resolution:** Optimized the tracking pipeline and utilized multi-threaded frame processing for efficient real-time execution.
- **Zone Synchronization:** Rendering multiple dynamic zones on live video feeds caused misalignment issues.
- **Resolution:** Implemented pixel-coordinate mapping with precise scaling logic on HTML canvas to ensure accurate overlay alignment.

2. Operational Challenges

- **Resource Management:** Balancing dataset preparation, model tuning, and Flask integration within milestone timelines.
- **Resolution:** Adopted a structured sprint-based approach, allocating fixed time blocks for testing, debugging, and documentation.
- **Deployment Issues:** Hosting heavy ML models on limited local servers affected initial deployment trials.
- **Resolution:** Compressed YOLO weights and used Render cloud hosting for stable deployment and API accessibility.

3. Communication Challenges

- **Feature Prioritization:** Initial ambiguity about admin controls and live analytics scope led to feature overlap.
- **Resolution:** Conducted periodic mentor consultations and established clear development hierarchies for dashboard, admin, and user modules.

8. Learnings & Skills Acquired

- **Technical Skills:** Practical experience in Python, Flask, OpenCV, YOLOv8, Deep SORT, SQLite, and Chart.js for real-time data visualization and analytics.
- **AI & Computer Vision:** Strengthened knowledge of detection, tracking, and zone-based analytics using deep learning pipelines.
- **Project Management:** Enhanced scheduling, documentation, and milestone tracking using agile-based task distribution.
- **Soft Skills:** Developed analytical problem-solving, technical communication, and collaboration through iterative review sessions and teamwork.

9. Testimonials from Experience

Working on the CrowdCount: People Counting Using Video Analytics project was a highly valuable and insightful experience. It provided me the opportunity to apply computer vision and AI-driven analytics to address real-world safety and monitoring challenges.

Key achievements include:

- Successfully building a real-time crowd counting system with live visualization and automated alerts.
- Gaining practical exposure to object detection, tracking algorithms, and Flask web integration.
- Enhancing my problem-solving and analytical thinking through model optimization and system debugging.
- Understanding the importance of collaboration, testing, and iterative refinement in developing production-ready AI systems.

Overall, this project significantly strengthened my confidence in designing and implementing **AI-based intelligent surveillance systems** for real-world applications.

10. Conclusion

The project was an enriching and transformative experience that bridged the gap between theoretical knowledge and real-world implementation. Developing the AI-powered CrowdCount System improved my technical proficiency in **computer vision, machine learning, and web-based visualization frameworks**.

- It also enhanced my understanding of end-to-end project development, from data handling and model deployment to user interface design and performance evaluation.
- This project has been a key milestone in my journey toward becoming a skilled AI engineer and full-stack computer vision developer, capable of building impactful, human-centered technology solutions.

11. Acknowledgements

I sincerely thank **Infosys Springboard** for providing the opportunity to work on this project.

I am deeply grateful to my mentor **Ms. Umme Asma S** for her constant guidance, technical insights, and constructive feedback throughout the internship.

I also thank my peers and collaborators for their support, teamwork, and encouragement, which made this project a rewarding experience.